UNIVERSITY OF LOUISVILLE

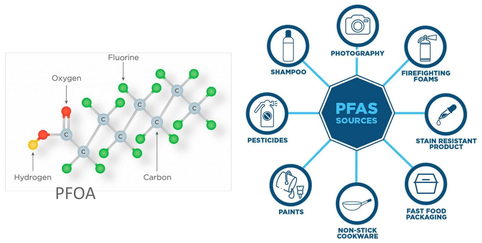
iGEM Louisville: Insights into Molecular Interactions—Predicting and Validating PFOA Binding to Human Proteins

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Abstract

PFAS (per- and polyfluoroalkyl substances) are chemicals used in the manufacturing of many products. They have been called "forever chemicals" for their inability to break down in our environment and have accumulated to an extent that at least 97% of Americans have traceable amounts in their blood (NHANES 2015). However, while the full spectrum of their effects is not entirely understood, they have been implicated in several cancers and liver conditions. Current detection methods like mass spectrometry and liquid or gas chromatography are expensive and inaccessible to the general public. Our project aims to develop a cost-effective protein-based detection system for PFAS. Using a reverse-screening approach across multiple protein-ligand databases, we identified candidate human proteins predicted to bind perfluorooctanoic acid (PFOA), a representative PFAS. Among these, we purified and characterized a key DNA repair protein is significantly destabilized in the presence of PFOA based on results from differential scanning fluorimetry (DSF).



Methods

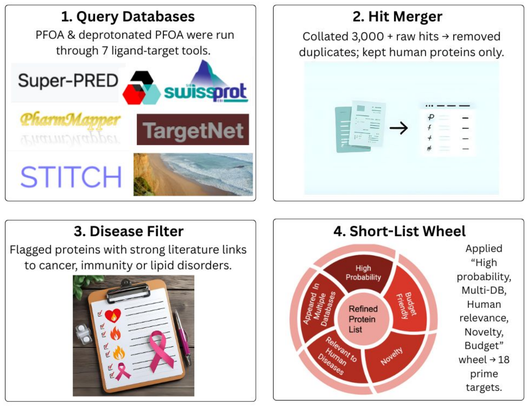
Molecular Dynamics

We used AMBER molecular dynamics simulations to investigate how our predicted proteins interact with PFOA. These simulations provide a detailed, atomic-level understanding of the putative binding and interaction with PFOA. Specifically, we aimed to identify: (1) the precise binding sites of PFOA on each protein; (2) the binding affinities involved; (3) the role of cofactors in modulating PFOA binding; and (4) the impact of protein oligomerization on these interactions. All simulations were carried out under neutral aqueous conditions with NaCl present, to mimic a physiological environment.

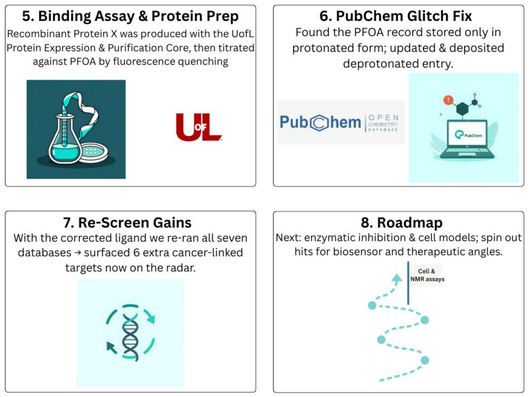
Reverse Screening for Identification of Putative Binding Partners

Reverse screening is a technique used to identify protein interactions. A target molecule (e.g., PFOA) is used to query a vast collection of human proteins, encompassing data from seven distinct protein databases.

Reverse Screening Pipeline



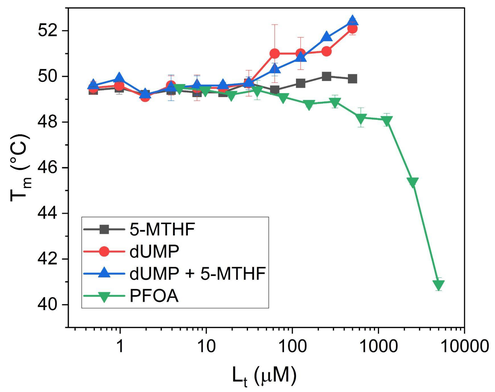
Validation & Refinement



Results

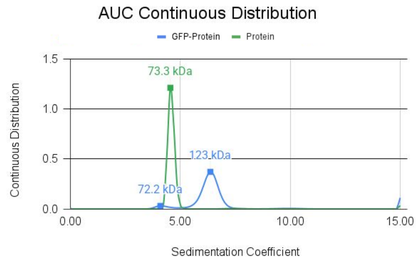
Differential Scanning Fluorimetry

The reverse screening identified multiple proteins to interact with PFOA. Five of the identified proteins were tested using DSF. Protein 1 and Protein 2 (names to be disclosed upon publication) were confirmed to interact with PFOA using DSF. Further DSF analysis was conducted with Protein 1 to assess the impact of increasing PFOA concentrations on its stability.

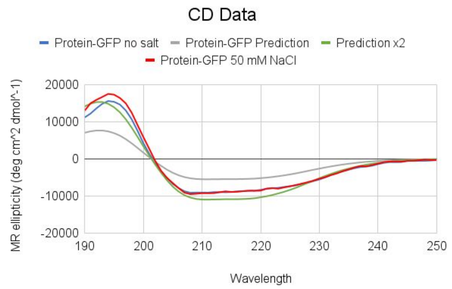


Protein Validation

We used Analytical Ultracentrifugation (AUC) and Circular Dichroism in order to investigate the folding and oligomerization of our proteins. We produced both the WT-protein and a GFP-Protein fusion. AUC results showed that each protein is within 4.0% and 8.4% of anticipated MW (mean absolute percentage error, MAPE), respectively, indicating that both proteins are folded, full-length, and form dimers.

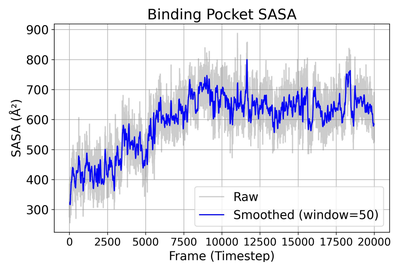


We used Circular Dichroism to check if the protein is folded properly. While the signal strength differed due to concentration differences, the overall shape matched well to calculated curves, indicating that the protein has the amount of beta and alpha content as predicted.

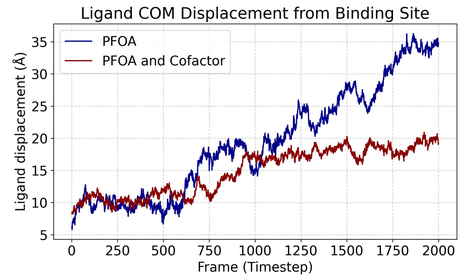


Molecular Dynamics

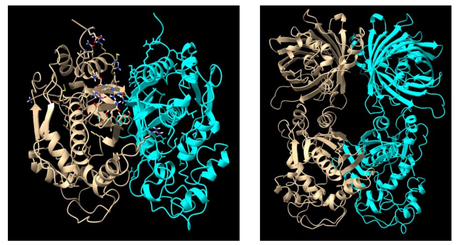
A simulation of PFOA in its predicted binding pocket showed a sustained pocket opening via a Solvent-Accessible Surface-Area (SASA) plot, suggesting an effect on enzyme kinetics.



We investigated if the presence of a cofactor affects the binding. The graph below demonstrates that in the presence of a cofactor, the displacement of PFOA significantly reduces, suggesting a stabilizing effect of PFOA's position relative to its binding site.



AlphaFold3 and MD Simulation: Protein Folding Analysis (colored by unit)



Protein dimer GFP-Protein dimer

Conclusions

- Identified, produced, and confirmed that PFAS destabilizes a cancer protein
- Identified protein is a widely known and important cancer target

Future Plans

- Utilize NMR to determine PFOA effects on enzyme kinetics
- Develop a mechanism for detection
- Run a DSF study with the newly reverse screened proteins

The iGEM Competition

iGEM is an annual synthetic biology competition in which student teams from all over the world design, build, and test biological systems using standardized genetic components known as BioBricks. These teams work together to solve real-world problems by applying biology, engineering, and computer science principles to develop innovative solutions. Our team received silver in the 2023 and gold in the 2024 iGEM competition!

Our GoFundMe ↴ Our 2024 Wiki ↴

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